

GroundVTS: Visual Token Sampling in Multimodal Large Language Models for Video Temporal Grounding

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Abstract

Video temporal grounding (VTG) is a critical task in video understanding and a key capability for extending video large language models (Vid-LLMs) to broader applications. However, existing Vid-LLMs rely on uniform frame sampling to extract video information, resulting in a sparse distribution of key frames and the loss of crucial temporal cues. To address this limitation, we propose Grounded Visual Token Sampling (GroundVTS), a Vid-LLM architecture that focuses on the most informative temporal segments. GroundVTS employs a fine-grained, query-guided mechanism to filter visual tokens before feeding them into the LLM, thereby preserving essential spatio-temporal information and maintaining temporal coherence. Furthermore, we introduce a progressive optimization strategy that enables the LLM to effectively adapt to the non-uniform distribution of visual features, enhancing its ability to model temporal dependencies and achieve precise video localization. We comprehensively evaluate GroundVTS on three standard VTG benchmarks, where it outperforms existing methods, achieving a 7.7-point improvement in mIoU for moment retrieval and 12.0-point improvement in mAP for highlight detection. Code is available at the [GroundVTS code repository](#).

1. Introduction

Understanding complex video content is fundamental to a wide range of applications, including video classification [4, 9, 11, 62], video captioning [5, 23, 63], video question-answering (VQA) [52, 56, 57], and video temporal grounding (VTG) [7, 18, 19]. Among these applications, VTG is a representative task for fine-grained temporal understanding: it aims to precisely identify the video segment corresponding to a given natural-language query [7].

Driven by the success of large-scale multimodal pretraining, numerous vision language models have achieved re-

markable progress in multimodal reasoning [8, 38, 40, 43, 64]. Meanwhile, rapid advances in large language models (LLMs) have opened new opportunities to incorporate high-level reasoning and instruction following into video understanding. When combined with visual encoders, LLMs provide a unified multimodal framework for interpreting complex temporal events and cross-modal relationships [42]. Building on this trend, many studies have developed video LLMs (Vid-LLMs) [22, 26, 32], achieving notable progress in general video reasoning. However, current Vid-LLMs still struggle with granular temporal understanding.

Prior efforts to enhance Vid-LLMs for temporal grounding have introduced query-conditioned attention [58], temporal boundary regressors [30], and temporal modeling modules [3]. Yet these approaches often overlook the sampling and representation of visual tokens before entering the LLM, which can be important for precise temporal reasoning. As shown in Figure 1(a), most existing approaches adopt uniform frame sampling, a dominant strategy in video understanding tasks. While this strategy provides consistent temporal coverage, it allocates the input budget evenly across time; consequently, key moments can be diluted or even missed, especially when query-relevant events are sparse. Recently, several studies [25, 46, 49] introduce query-guided frame sampling at the video input stage by attaching an external multimodal encoder (e.g., CLIP [38] or a captioning model) to compute cross-modal similarity for frame selection (Figure 1(b)). However, they perform coarse-grained filtering and depend on auxiliary encoders, limiting localization precision and adaptability for VTG.

To address these limitations, we propose GroundVTS, a Vid-LLM architecture that performs query-guided visual token sampling at a finer granularity. As illustrated in Figure 1(c), GroundVTS introduces a Visual Token Sampling (VTS) module that operates after the visual encoder and multimodal projection layers. Unlike the coarse selection used in prior work, VTS selectively retains visual tokens based on token-level similarity to the textual query, yielding a non-uniform token distribution across both temporal and spatial dimensions. This helps the Vid-LLM focus on

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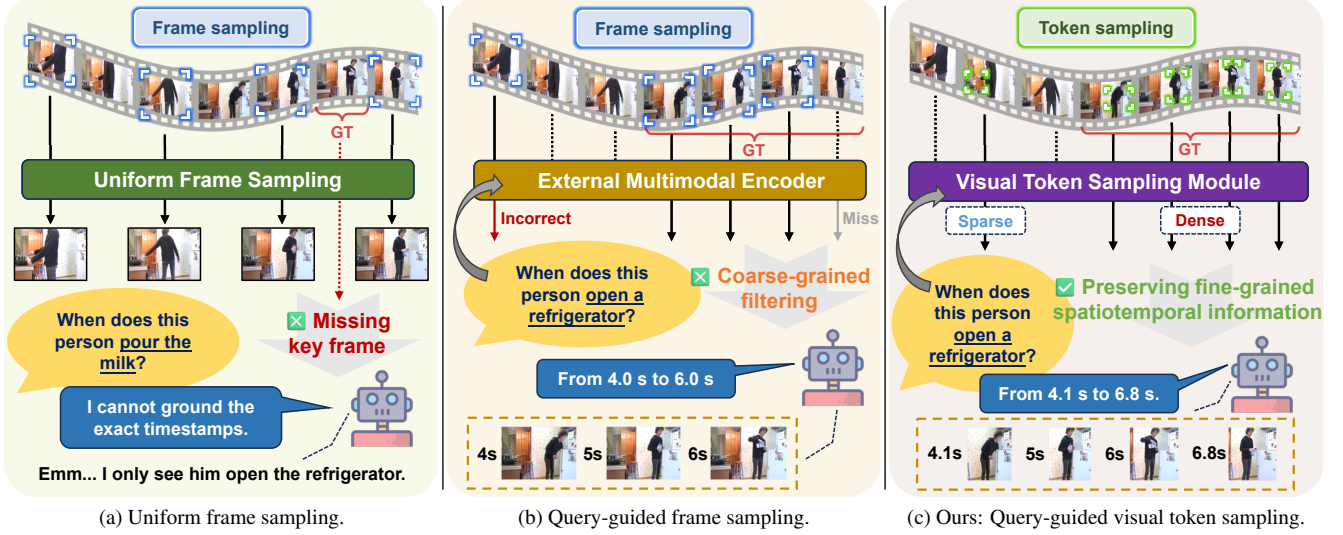


Figure 1. **Comparison of sampling strategies for Vid-LLMs.** (a) Uniform sampling distributes attention evenly across frames, often missing query-relevant moments; (b) existing query-guided frame sampling relies on external encoders for coarse frame selection, limiting temporal understanding; and (c) our query-guided VTS adaptively selects query-relevant visual tokens within Vid-LLMs while maintaining temporal coherence, enabling efficient and precise video grounding.

crucial cues while suppressing irrelevant visual content, enabling more precise and robust temporal grounding.

Our main contributions are summarized as follows:

- We propose GroundVTS, a Vid-LLM framework with a query-guided VTS strategy that adaptively preserves critical spatio-temporal cues to sharpen temporal localization for VTG.
- We introduce a progressive optimization strategy that integrates the VTS module into existing Vid-LLM architectures, enabling effective adaptation to non-uniform visual token distributions while ensuring stable training.
- We conduct extensive experiments across multiple VTG benchmarks under varying visual token densities and model architectures, demonstrating the effectiveness and generality of GroundVTS.

2. Related Work

Video large language models (Vid-LLMs) are typically implemented by connecting visual encoders to LLMs through projection layers or multimodal adapters. Recent studies have focused on enhancing the global video understanding capabilities [22, 26, 32, 47, 49]. While these models achieve strong performance, they still struggle to capture granular temporal structures [29]. GroundVTS instead emphasizes fine-grained temporal perception, thereby improving the ability of Vid-LLMs to handle challenges of VTG.

Video temporal grounding (VTG) is a fundamental video understanding task that aims to accurately localize the timestamps of events within a video [27]. Classical expert models formulate this as a cross-modal matching prob-

lem, employing proposal-based or regression-based architectures to improve grounding precision [13, 28, 54, 59]. These models are typically task-specific and require dataset-level fine-tuning [45]. Recent studies [6, 10, 14, 15, 35, 51] integrate VTG capabilities into Vid-LLMs, leveraging their strong reasoning and generalization strengths to unify diverse video understanding tasks within a single framework. To support VTG within this unified setting, many methods introduce explicit time representations (*e.g.*, time tokens/position embeddings) or frame indexing to better model timestamps [10, 51]. However, these approaches primarily focus on temporal encoding or data adaptation; the sampling and representation of visual tokens before entering the LLM have received comparatively less attention. GroundVTS addresses this gap by introducing a unified Vid-LLM framework with an effective token sampling mechanism that substantially improves temporal understanding.

Visual token compression and selection. In VTG and VQA, not all visual tokens contribute equally to answering a given query, and many contain redundant or irrelevant information that can distract the model from the relevant evidence [34]. This has motivated growing interest in token compression, pruning, and adaptive selection for Vid-LLMs [1, 16, 41]. However, most existing approaches rely on query-agnostic or saliency-based token reduction, often preserving visually prominent yet semantically irrelevant tokens, which can limit temporal reasoning and grounding precision. Recent token compression methods further deliver strong efficiency gains and improve scalability for long-video understanding [24, 31, 61]. Their impli-

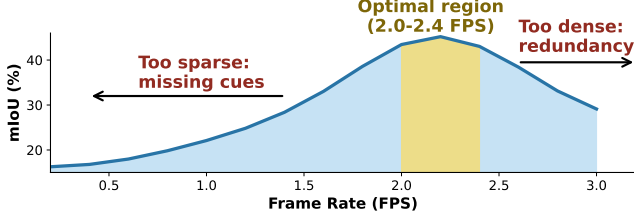


Figure 2. Frame rate sensitivity of Qwen2.5VL-7B. Similar trends hold for InternVL3.5 (illustrated in the supplementary material).

cations for VTG—where fine-grained temporal localization is central—remain less explored. In contrast, our GroundVTS dynamically samples query-relevant tokens within the Vid-LLM via token–query relevance, preserving spatio-temporal cues without external preprocessing.

3. Method

We propose GroundVTS, a Vid-LLM architecture designed to enhance VTG performance through adaptive and efficient visual token utilization. Sec. 3.1 analyzes the sensitivity of VTG performance to frame density, which highlights the necessity of adaptive token sampling, motivating our design. Sec. 3.2 presents the overall architecture of GroundVTS, followed by a detailed explanation of the Visual Token Sampling (VTS) module in Sec. 3.3. Finally, Sec. 3.4 outlines our progressive optimization strategy, which allows VTS to be seamlessly integrated into existing Vid-LLMs.

3.1. Analysis of Frame Rate Sensitivity

Before introducing our sampling strategy, we first examine how the density of visual tokens inherently influences VTG performance. To this end, we evaluate the pretrained Qwen2.5VL-7B [3] model on the Charades-STA [7] dataset under varying frame rates from 0.2 to 3.0 frames per second (FPS), while keeping all other settings fixed.

As shown in Figure 2, VTG performance demonstrates a clear non-linear dependency on frame rate. When the frame rate is low (< 1.0 FPS), the model lacks sufficient temporal cues, resulting in degraded mIoU. As the sampling rate increases, performance improves steadily and reaches its peak around 2.0–2.4 FPS, where mIoU attains 47.8%. However, further increasing the frame rate beyond this range leads to a sharp performance drop, indicating that redundant visual tokens dilute key temporal signals and hinder accuracy.

This observation supports our core hypothesis that the density and relevance of visual tokens critically influence VTG performance. Consequently, an adaptive token sampling mechanism is essential for achieving both accuracy and efficiency in temporal grounding, motivating the design of our GroundVTS framework.

3.2. GroundVTS

The GroundVTS framework, illustrated in Figure 3(a), incorporates a query-guided visual token sampling module to enable efficient and fine-grained temporal grounding. Unlike prior methods that rely on additional preprocessing [49], GroundVTS integrates the sampling process directly into the Vid-LLM pipeline, allowing the model to dynamically focus on temporally and semantically relevant segments conditioned on the input query.

Specifically, given an input video after standard temporal downsampling, $\mathcal{V} = \{F_t\}_{t=1}^T$, and a text query, the query is first tokenized into a sequence of embeddings, $Q = [q_1, q_2, \dots, q_{N_t}] \in \mathbb{R}^{N_t \times D}$, using a pretrained language tokenizer. Here, T denotes the number of frames, N_t is the number of text tokens, and D is the text token embedding dimension. Each video frame F_t is divided into spatial patches and encoded by a pretrained vision encoder to produce dense spatio-temporal features, $H_v = [h_1, h_2, \dots, h_{N_v}] \in \mathbb{R}^{N_v \times D_v}$, where N_v denotes the total number of visual tokens and D_v denotes the feature dimension. Then, a multimodal projector (implemented as a multilayer perceptron) maps these features to a shared embedding space: $V = [v_1, v_2, \dots, v_{N_v}] \in \mathbb{R}^{N_v \times D}$, where each $v_i \in \mathbb{R}^D$ is the projected embedding of the i -th visual token, ensuring that visual embeddings are dimensionally aligned with the text token embeddings Q .

Conditioned on the query embeddings Q , the VTS module (which will be introduced in Sec. 3.3) evaluates the relevance of each visual token to the query and outputs a compact subset of visual token embeddings via a weighted differentiable top- K selection mechanism:

$$\tilde{V} = \text{VTS}(V, Q) = [\tilde{v}_1, \tilde{v}_2, \dots, \tilde{v}_K], \quad (1)$$

where K is the number of tokens selected by VTS. The resulting $\tilde{V}$ forms a non-uniform, query-guided distribution of visual tokens that allocates denser sampling around relevant moments and sparser coverage elsewhere.

To preserve temporal coherence under non-uniform sampling, we reuse the original positional encodings from dense sampling (masking out only those of unselected visual tokens), ensuring that the selected tokens remain temporally aligned with the input video.

Finally, the sampled visual tokens and query embeddings are concatenated to form a multimodal sequence and processed by the LLM for joint reasoning and generation.

3.3. Visual Token Sampling

The VTS module is the core of GroundVTS, responsible for dynamically selecting the most informative visual tokens under the guidance of the textual query. As illustrated in Figure 3(b), VTS consists of the following two main steps.

Query-Guided Token Scoring. Given the projected visual embeddings, $V = \{v_i\}_{i=1}^{N_v}$, and the tokenized query

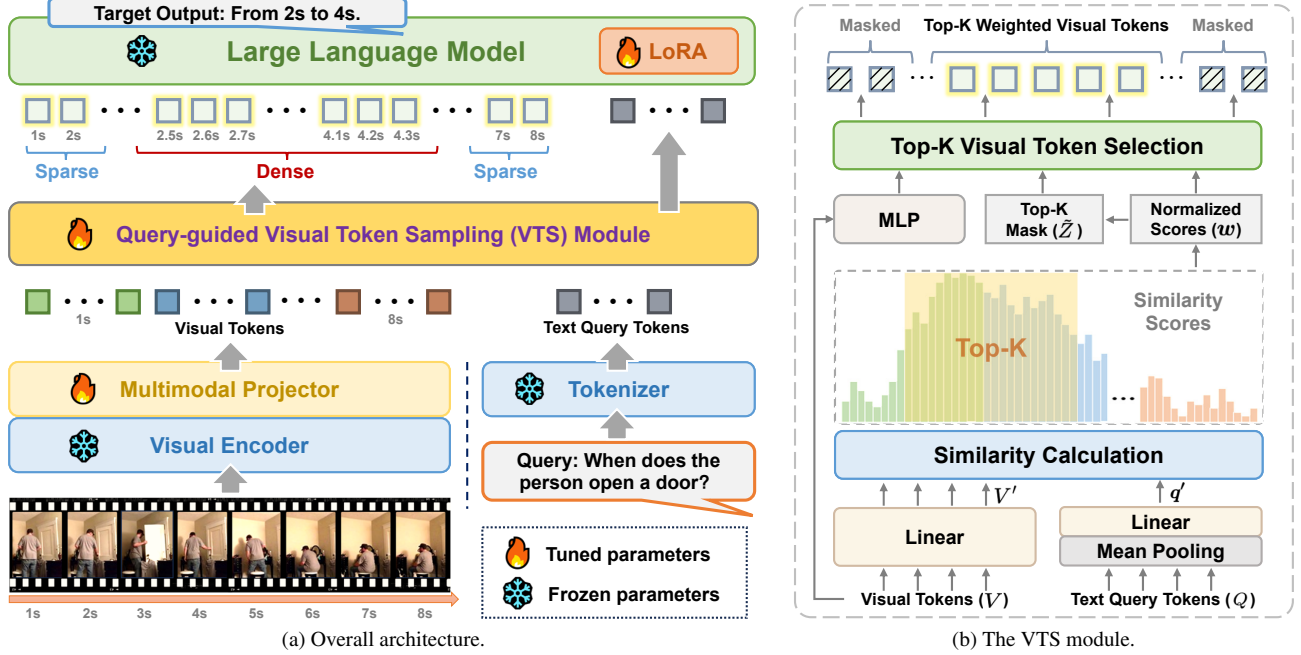


Figure 3. **Overview of the proposed GroundVTS framework.** (a) GroundVTS integrates a query-guided VTS module into the Vid-LLM pipeline, enabling adaptive selection of query-relevant tokens; (b) The VTS module computes token-query similarity scores and performs weighted differentiable top- K sampling to retain the most informative tokens, supporting efficient and precise video temporal grounding.

embeddings, $Q = \{q_j\}_{j=1}^{N_t}$, to estimate the relevance between each visual token and the query, both sets of embeddings are first projected into lower-dimensional subspaces of dimension D_r through linear projections:

$$V' = W_v V, \mathbf{q}' = W_q \text{Pool}(Q), W_v, W_q \in \mathbb{R}^{D \times D_r}. \quad (2)$$

Here, W_v and W_q are trainable projection matrices, and $\text{Pool}(\cdot)$ denotes mean pooling over text token embeddings.

Then, the relevance distribution is obtained through a softmax function applied to the temperature-scaled dot products between each pair of embeddings in V' and $\mathbf{q}'$:

$$\mathbf{w} = \text{softmax}(V' \mathbf{q}'^\top / \tau), \quad (3)$$

where τ is a temperature hyperparameter controlling the sharpness of the distribution. This formulation can be interpreted as an attention mechanism, where the weight assigned to each visual token reflects both its alignment with the query and its relative importance within the sequence. The resulting weight vector, $\mathbf{w} = [w_1, w_2, \dots, w_{N_v}]^\top$, effectively emphasizes semantically relevant visual tokens while attenuating less informative ones.

Differentiable Top- K Selection. To efficiently retain the most relevant visual information, we select the top- K visual tokens based on $\mathbf{w}$. The number of selected tokens is adaptively determined by the ratio $\rho \in (0, 1]$, where $K = \lceil \rho \cdot N_v \rceil$. Let $\mathcal{I}_K$ denote the indices of the top- K tokens.

Since hard top- K selection is non-differentiable, we employ a Straight-Through Estimator (STE) with Gumbel-Softmax relaxation [17] to enable end-to-end training. Specifically, we add Gumbel noise $g_i \sim \text{Gumbel}(0, 1)$ to the log-probabilities and compute a differentiable approximation of the discrete selection mask:

$$z_i = \frac{\exp((\log w_i + g_i) / \tau_g)}{\sum_{j=1}^{N_v} \exp((\log w_j + g_j) / \tau_g)}, \quad (4)$$

where τ_g is a Gumbel temperature controlling the smoothness of the relaxation. During the forward pass, a hard top- K operator is applied:

$$z_i^{\text{hard}} = \begin{cases} 1, & \text{if } i \in \mathcal{I}_K, \\ 0, & \text{otherwise,} \end{cases} \quad (5)$$

while the backward propagation flows through the continuous relaxation z_i . To this end, the STE is implemented as the following, combining the hard and soft representations:

$$\tilde{z}_i = z_i^{\text{hard}} + z_i - \text{stopgrad}(z_i), \quad (6)$$

where $\text{stopgrad}(\cdot)$ denotes gradient detachment.

The final visual representations are then obtained via weighted differentiable top- K masking: the relevance weights of the selected top- K tokens are re-normalized to sum to one and serve as the non-zero elements of the final

mask, while all non-top- K positions remain zero. Mathematically, this is formulated as:

$$\tilde{v}_i = \hat{w}_i \cdot \text{MLP}(\mathbf{v}_i), \quad \hat{w}_i = \frac{\exp(w_i/\tau') \cdot \tilde{z}_i}{\sum_{j=1}^{N_v} \exp(w_j/\tau') \cdot \tilde{z}_j}, \quad (7)$$

where MLP indicates a multilayer perceptron and τ' represents the temperature hyperparameter. This strategy ensures that token selection remains query-aware and trainable.

3.4. Progressive Optimization Strategy

To ensure stable convergence and effective cross-modal adaptation, GroundVTS is optimized through a three-stage progressive training strategy. Each stage focuses on a specific learning objective, allowing the model to gradually develop non-uniform visual token sampling and query-conditioned reasoning capabilities while maintaining the stability of the underlying Vid-LLM.

Stage 1: VTS Warm-up. The first stage aims to initialize the query-guided visual token sampling process. Since the VTS module dynamically samples tokens based on query relevance, jointly training it with the LLM from scratch can lead to unstable gradients and inconsistent selection behavior. To mitigate this, we train only the parameters of the VTS module while freezing all other components. This warm-up phase enables VTS to learn robust visual token-query relevance estimation and stable visual importance prediction before engaging in joint optimization.

Stage 2: Joint LoRA Adaptation. After the VTS has learned stable sampling behavior, the second stage focuses on aligning the non-uniform visual token distributions with textual semantics. We fine-tune the LLM using Low-Rank Adaptation (LoRA) [12] while jointly updating the VTS and the multimodal projector. This stage leverages a large-scale multimodal dataset LLaVA-Video-178K [60], exposing the model to diverse temporal structures and cross-modal reasoning scenarios. Through this joint optimization, the model learns to effectively interpret and reason over query-guided, non-uniform visual token sequences, enabling fine-grained and efficient temporal reasoning within the LLM.

Stage 3: Grounding Fine-tuning. The final stage aims to refine the model’s temporal grounding and reasoning ability for real-world VTG tasks. We fine-tune GroundVTS on a newly curated dataset, Grounding-FT (see Sec. 4.1), which aggregates training samples from multiple temporal grounding datasets. Each instance in Grounding-FT follows the same instruction-style QA format as in previous stages, and is used to guide the model to predict temporal boundaries or highlight relevant events. This unified instruction-based interface enables GroundVTS to perform temporal grounding and descriptive reasoning jointly within a single generative process. Throughout this stage, the mod-

ule freezing configuration follows that of Stage 2, ensuring smooth and stable fine-tuning.

4. Experiments

4.1. Experimental Setup

Dataset Preparation. We employ two datasets in the training pipeline: (a) **LLaVA-Video-178K** [60], a large-scale video dataset that provides diverse multimodal supervision across tasks such as video captioning and VQA for pre-training and alignment; and (b) **Grounding-FT**, a curated dataset we construct for VTG tasks. To adapt VTG supervision to the natural language input-output format of LLMs, we design a set of instruction templates with diverse linguistic expressions and combine them with temporal grounding queries to form QA-style training pairs. Grounding-FT is derived from multiple VTG training splits covering both moment retrieval and highlight detection tasks [7, 18, 19], and contains a total of 70K annotated video-query pairs. Dataset details are provided in the supplementary material. LLaVA-Video-178K is used in the first two training stages, while Grounding-FT is employed in the third stage.

Evaluation. We evaluate our model on two representative VTG tasks, namely moment retrieval (MR) [4, 7] and highlight detection (HD) [19]. The MR task aims to identify the start and end timestamps of the video segment corresponding to a given natural language query. Following standard practice, we conduct evaluation on Charades-STA [7], ActivityNet-Captions [4], and QVHighlights [19], using mean intersection-over-union (mIoU) and Recall@1 (R1@t) at thresholds $t \in \{0.3, 0.5, 0.7\}$ [21, 37, 51]. The HD task requires the model to output all salient moments relevant to the query in the video together with their corresponding relevance scores. We use QVHighlights [19] for evaluation and adopt mean average precision (mAP) and the hit ratio of the highest-scored clip (Hit@1) as metrics [10, 27, 39].

Implementation Details. We construct two model variants, GroundVTS-Q and GroundVTS-I, built upon Qwen2.5VL-7B [3] and InternVL3.5-8B [44], respectively. Both models are trained using the three-stage strategy described in Sec. 3.4, where stages 1–3 are trained for 1, 2, and 3 epochs, respectively, with learning rates of 1×10^{-5} , 2×10^{-4} , and 1×10^{-4} . The two base models differ in their intrinsic video sampling paradigms. QwenVL employs a fixed frame-rate strategy, uniformly sampling frames over time, whereas InternVL adopts a fixed frame-count strategy, representing each video with a constant number of frames regardless of duration. During training, GroundVTS-Q uses a frame rate of 2 FPS, while GroundVTS-I samples 16 frames per video. For the VTS module, the hidden dimension D_r is set to 512 for GroundVTS-Q and 128 for GroundVTS-I. The visual token sampling ratio is fixed at $\rho = 0.5$. Additional training settings are detailed in the supplementary material.

Table 1. Comparison with state-of-the-art methods on Charades-STA and ActivityNet-Captions test splits.

Method	Charades-STA				ActivityNet-Captions			
	R1@.3	R1@.5	R1@.7	mIoU	R1@.3	R1@.5	R1@.7	mIoU
LLaVA-OV[20] arXiv' 24	28.8	16.6	5.9	19.3	20.2	8.6	2.2	13.5
TimeChat[39] CVPR' 24	47.7	22.9	12.5	30.6	30.2	16.9	8.2	21.8
VTimeLLM[14] CVPR' 24	51.0	27.5	11.4	31.2	44.0	27.8	14.3	30.4
Momentor[35] ICML' 24	42.9	23.0	12.4	29.3	42.6	26.6	11.6	28.5
HawkEye[48] arXiv' 24	50.6	31.4	14.5	33.7	49.1	29.3	10.7	32.7
ChatVTG[37] CVPR' 24	52.7	33.0	15.9	34.9	40.7	22.5	9.4	27.2
NumPro[51] CVPR' 25	<u>63.8</u>	42.0	20.6	41.4	55.6	37.5	<u>20.6</u>	38.8
LLaVA-ST[21] CVPR' 25	63.1	<u>44.8</u>	23.4	<u>42.4</u>	–	–	–	–
Qwen2.5VL-7B	34.2	18.8	8.6	22.1	25.3	11.5	4.4	17.1
Qwen2.5VL-7B-G	45.2	32.7	18.7	31.7	40.6	23.9	9.9	26.7
GroundVTS-Q (ours)	71.5 ($\uparrow 26.3$)	57.5 ($\uparrow 24.8$)	34.2 ($\uparrow 15.5$)	50.1 ($\uparrow 18.4$)	<u>51.3</u> ($\uparrow 10.7$)	<u>33.6</u> ($\uparrow 9.7$)	21.4 ($\uparrow 11.5$)	<u>36.0</u> ($\uparrow 9.3$)
InternVL3.5-8B	35.5	25.7	13.2	24.6	22.1	12.0	5.6	15.8
InternVL3.5-8B-G	59.5	42.0	20.2	39.4	35.9	20.6	9.0	24.5
GroundVTS-I (ours)	61.2($\uparrow 1.7$)	44.2($\uparrow 2.2$)	<u>23.7</u> ($\uparrow 3.5$)	41.6($\uparrow 2.2$)	37.9($\uparrow 2.0$)	22.4($\uparrow 1.8$)	10.3($\uparrow 1.3$)	25.7($\uparrow 1.2$)

Bold denotes the best, underlined denotes the second-best. “-G” denotes supervised fine-tuning on the Grounding-FT dataset. “-Q” and “-I” denote our proposed models based on Qwen2.5VL-7B [3] and InternVL3.5-8B [44], respectively. $\uparrow$ indicates improvement over the corresponding “-G” baseline.

Table 2. Comparison with state-of-the-art methods on QVHighlights validation split.

Method	MR		HD	
	R1@.5	R1@.7	mAP	Hit@1
SeViLA ^o [55]	54.5	36.5	–	–
UniVTG ^o [27]	<u>58.9</u>	40.9	27.0	55.3
VTG-LLM[10]	–	–	16.5	33.5
TimeChat[39]	–	–	14.5	23.9
NumPro[51]	–	–	40.5	<u>70.7</u>
Qwen2.5VL-7B	8.7	2.4	24.9	0.6
Qwen2.5VL-7B-G	11.0	4.3	34.4	44.5
GroundVTS-Q	23.6($\uparrow 12.6$)	12.3($\uparrow 8.0$)	<u>35.7</u> ($\uparrow 1.3$)	58.8($\uparrow 14.3$)
InternVL3.5-8B	8.7	3.7	24.8	0.32
InternVL3.5-8B-G	31.8	15.0	31.9	39.8
GroundVTS-I	63.6 ($\uparrow 31.8$)	<u>40.7</u> ($\uparrow 25.7$)	52.5 ($\uparrow 20.6$)	88.4 ($\uparrow 48.6$)

^o indicates classical expert models; other notations follow Table 1.

4.2. Main Results

Moment Retrieval. As summarized in Tables 1 and 2, our proposed GroundVTS consistently outperforms existing state-of-the-art methods on multiple VTG benchmarks. On Charades-STA, GroundVTS-Q substantially outperforms the fine-tuned Qwen2.5VL-7B baseline, achieving gains of 24.8 points in R1@0.5 and 18.4 points in mIoU, reaching 57.5 R1@0.5 and 50.1 mIoU. On ActivityNet-Captions, GroundVTS-Q improves R1@0.5 by 9.7 points and mIoU by 9.3 points, further confirming the effectiveness of our sampling approach for VTG. Building upon InternVL3.5-8B, GroundVTS-I also shows stable improvements (e.g., +3.5 in R1@0.7 on Charades-STA), validating the generality of our approach across diverse Vid-LLM architectures.

Table 3. Comparison with state-of-the-art methods on NExT-GQA test splits.

Model	mIoU	mIoP	IoU@.5	IoP@.5	Acc@GQA
TOGA ^o [33]	<u>24.4</u>	40.5	21.1	40.6	24.6
VideoStreaming [36]	19.3	32.2	13.3	31.0	17.8
GroundVTS-Q	25.8	<u>37.4</u>	<u>20.4</u>	<u>35.4</u>	<u>23.2</u>
GroundVTS-I	16.7	26.5	11.9	24.3	18.5

On QVHighlights, GroundVTS-I attains 63.6 in R1@0.5 and 40.7 in R1@0.7 for moment retrieval, comparable to specialized methods such as UniVTG [27].

Highlight Detection. As shown in Table 2, GroundVTS-I significantly outperforms InternVL3.5-8B-G, improving mAP and Hit@1 by 20.6 and 48.6 points, respectively, to 52.5 and 88.4. It also surpasses strong methods using frame indices as auxiliary inputs, such as NumPro [51], suggesting better sensitivity to key moments in highlight detection.

Out-of-Distribution Evaluation. To further assess the effectiveness and generality of our method under task shift, we evaluate our models *as-is* on grounded video question answering with NExT-GQA [53], without any further training; results are shown in Table 3. GroundVTS-Q achieves the highest mIoU and remains competitive on other metrics, despite not being specifically designed or trained for this task. Moreover, the supplementary material reports two additional *as-is* evaluations: DiDeMo [2] for out-of-distribution moment retrieval, and LongVideoBench [50] for transfer to a new long-video understanding task; on both benchmarks, our models either outperform or remain competitive with recent state-of-the-art methods.

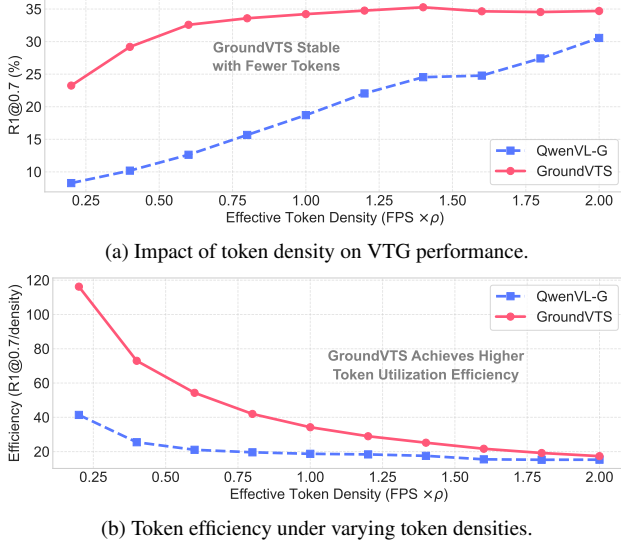


Figure 4. Comparison between GroundVTS-Q and Qwen2.5VL-7B-G (denoted as QwenVL-G) under varying token densities.

4.3. Effect of Visual Token Density

To verify the robustness of our GroundVTS with respect to visual token density, we conduct an analysis on the Charades-STA test split, comparing GroundVTS-Q with its fine-tuned base model, Qwen2.5VL-7B-G (abbreviated as QwenVL-G). We adjust the sampling ratio ρ from 0.1 to 1.0 in increments of 0.1, while keeping the dense sampling frame rate fixed at 2 FPS, to control the number of visual tokens involved in LLM inference. For a fair comparison, QwenVL-G continues to use uniform frame sampling, with the frame rate varied from 0.2 to 2 FPS in increments of 0.2.

As shown in Figure 4(a), the horizontal axis represents effective token density ($\text{FPS} \times \rho$), and the vertical axis reports grounding accuracy in terms of R1@0.7. As the token density decreases, QwenVL-G degrades markedly, indicating a strong dependence on dense temporal sampling. In contrast, GroundVTS remains much more stable across the full density range, maintaining high accuracy even in sparse settings. With only half the token budget ($\text{FPS} \times \rho = 1.0$), GroundVTS achieves 34.2 R1@0.7, already surpassing QwenVL-G at full density (30.5 R1@0.7). Even under a more aggressive reduction ($\text{FPS} \times \rho = 0.4$), GroundVTS still attains 29.2 R1@0.7, exceeding QwenVL-G by 19.0 points. These results highlight the strong token efficiency and robustness of GroundVTS under sparse sampling.

Figure 4(b) illustrates token efficiency, defined as R1@0.7 divided by effective token density. When fewer visual tokens are available, GroundVTS-Q maintains higher efficiency than QwenVL-G, indicating more effective use of limited visual information. Exact values are provided in the supplementary material.

Table 4. Ablation of different training-stage combinations for GroundVTS-Q on Charades-STA test split.

Stage	R1@0.3	R1@0.5	R1@0.7	mIoU
base [‡]	34.2	18.8	8.6	22.1
None [§]	8.6(↓25.6)	5.0(↓13.8)	1.9(↓6.7)	5.6(↓16.5)
1	31.2(↓3.0)	20.5(↑1.7)	10.0(↑1.4)	20.9(↓1.2)
1, 2	45.8(↑11.6)	28.8(↑10.0)	13.2(↑4.6)	30.1(↑8.0)
1, 3	49.1(↑14.9)	32.5(↑13.7)	15.2(↑6.6)	32.4(↑10.3)
2, 3	69.4(↑35.2)	53.0(↑34.2)	30.5(↑21.9)	47.4(↑25.3)
1, 2, 3	71.5(↑37.3)	57.5(↑38.7)	34.2(↑25.6)	50.1(↑28.0)

[‡] Arrowed values indicate absolute changes relative to the base model (Qwen2.5VL-7B). [§] “None” uses a randomly initialized VTS module.

4.4. Effect of the Progressive Optimization Strategy

Table 4 summarizes the effect of different training stages for GroundVTS-Q. Using an untrained VTS module (the “None” setting) causes a sharp drop across all metrics, showing that query-conditioned token sampling must be properly learned. Stage 1 (VTS Warm-up) largely recovers the base-model performance, indicating that VTS can be integrated without disrupting the original pipeline. Adding Stage 2 (Joint LoRA Adaptation) further improves performance, bringing gains of +11.6 in R1@0.3 and +8.0 in mIoU over the base model. Adding Stage 3 (Grounding Fine-tuning) yields the best results, reaching 71.5/57.5/34.2 R1@0.3/0.5/0.7 and 50.1 mIoU. The (1, 3) and (2, 3) variants remain below the full setting, confirming the importance of Stage 2 for large-scale adaptation to non-uniform token distributions and Stage 1 for stable initialization.

4.5. Ablation Study

We conduct ablation experiments on two key components of GroundVTS in Table 5: the visual token sampling strategy and the positional encoding used for temporal reasoning. All variants are evaluated under a matched token budget, equivalent to $\text{FPS} = 2.0$ and $\rho = 0.5$.

Sampling Strategy. We compare our query-guided token-level sampling with three alternatives: (a) *Uniform sampling*, implemented by evaluating Qwen2.5VL-7B-G at 1.0 FPS; (b) *Random sampling*, where 50% of visual tokens are randomly discarded; and (c) *Frame-level query selection*, where visual tokens within each frame are average-pooled to estimate frame-query relevance, and the top 50% frames are retained with all their tokens. Both the token-level and frame-level variants are trained with the same three-stage procedure, while the random variant is initialized from the token-level model after Stages 1 and 2 and trained only in Stage 3 with random dropping.

As shown in Table 5, our token-level VTS achieves the best performance on both datasets. On Charades-STA, it improves mIoU over frame-level selection by 8.5 points

Table 5. Ablation on sampling strategies and positional encoding (PE) in GroundVTS.

VTS	PE	Sampling Methods	Charades-STA				ActivityNet-Captions			
			R1@0.3	R1@0.5	R1@0.7	mIoU	R1@0.3	R1@0.5	R1@0.7	mIoU
✓	✓	Token-Level	71.5	57.5	34.2	50.1	51.3	33.6	21.4	36.0
✓	✓	Frame-Level	<u>61.7</u>	<u>44.9</u>	<u>23.3</u>	<u>41.6</u>	<u>43.7</u>	<u>27.5</u>	<u>15.0</u>	<u>30.7</u>
–	✓	Uniform	42.6	28.5	15.0	29.3	36.1	19.5	7.5	23.4
–	✓	Random	54.9	35.0	16.3	35.7	40.3	23.4	12.1	27.7
✓	–	Token-Level	15.1	7.0	2.7	9.5	22.2	11.2	5.2	16.3

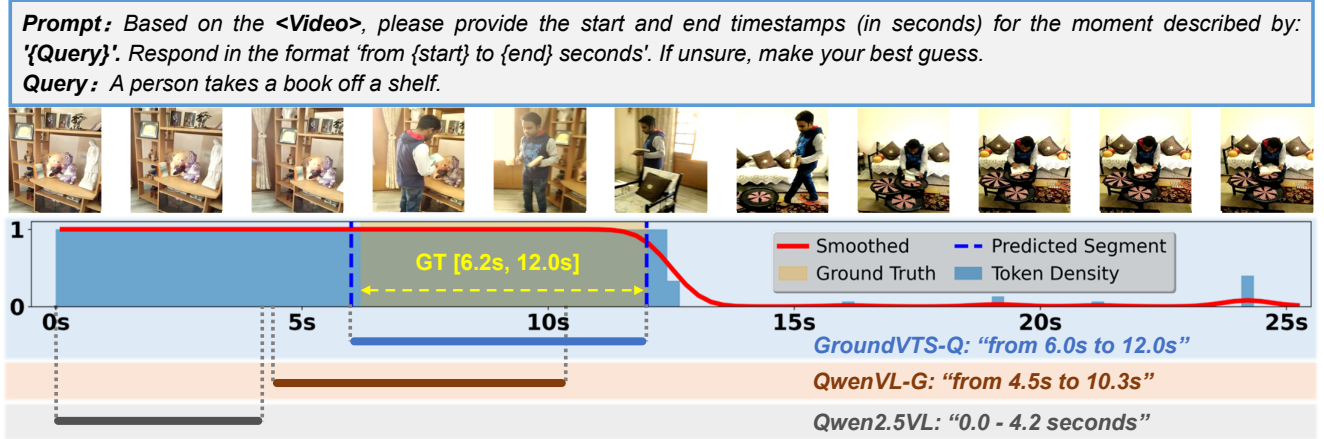


Figure 5. Qualitative comparison of temporal grounding predictions among GroundVTS-Q, Qwen2.5VL-7B-G, and Qwen2.5VL-7B.

(50.1 vs. 41.6) and reaches 57.5 R1@0.5. On ActivityNet-Captions, it also performs best, improving mIoU by 5.3 points over the frame-level variant (36.0 vs. 30.7). By contrast, both uniform and random sampling degrade performance, confirming the importance of query-guided fine-grained sampling for temporal grounding; random sampling nevertheless outperforms uniform sampling, possibly because it acts as data augmentation.

Effect of Positional Encoding. To assess the role of positional encoding, we remove position embeddings from GroundVTS while keeping training and inference settings fixed. As shown in Table 5, performance collapses on both datasets. On Charades-STA, mIoU drops from 50.1 to 9.5 and R1@0.5 from 57.5 to 7.0, with similarly severe degradation on ActivityNet-Captions. These results confirm the importance of temporal positional information in GroundVTS, and validate our design choice of retaining the original relative positional embeddings for the selected tokens.

Additional ablations on training data and relevance estimation are provided in the supplementary material.

4.6. Qualitative Study

Figure 5 shows a qualitative comparison on a Charades-STA example with the query “a person takes a book off a shelf.” The red curve denotes the normalized token density produced by VTS, with higher values indicating stronger

query relevance. GroundVTS-Q assigns most tokens to the early part of the video (roughly 0–13 s), which fully covers the ground-truth interval (6.2–12.0 s), while suppressing nearly all tokens in later frames. Based on these sampled tokens, GroundVTS-Q predicts 6.0–12.0 s, closely matching the ground truth. In contrast, Qwen2.5VL-7B-G predicts an earlier and less precise segment (4.5–10.3 s), while the base Qwen2.5VL-7B misses the target moment entirely. This shows that VTS focuses on relevant temporal regions for grounding. More results are provided in the supplement.

5. Conclusion

In this paper, we present GroundVTS, a query-guided visual token sampling framework for video temporal grounding. Its core module, VTS, can be seamlessly integrated into mainstream Vid-LLMs via a progressive optimization strategy to better capture fine-grained temporal cues. Experiments show that GroundVTS consistently improves instruction-tuned base models and outperforms recent state-of-the-art methods. Further analyses confirm that GroundVTS improves token utilization and maintains prediction stability across varying input densities.

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